Abhinav

M.Tech in AI for Sustainability (Final Year)
Indian Institute of Technology, Kanpur

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Education

Year	Degree/Certificate	Institute	CPI/%
2025–Present	M.Tech (AI for Sustainability)	Indian Institute of Technology, Kanpur	9.38/10
2021–2025	B.S. (Data Science and Engineering)	IISER Bhopal	7.54/10
2021	CBSE (Class XII)	BD DAV Dharamshala, H.P	88.4%
2019	CBSE (Class X)	BD DAV Dharamshala, H.P	83.8%

Research Experience

- **AI-Physics Hybrid Modelling of Land–Atmosphere Interactions**(M.Tech Thesis) *Guide: Dr. Sarosh Ghausi,(Ongoing)*
  - Benchmarked **memory-aware and memory-free ML models** (Linear Regression, Random Forest, PyTorch LSTM, and XGBoost) to estimate land water stress (f_w) across **74 global FLUXNET sites**; **14-day lag XGBoost** achieved the best performance with **R^2 = 0.601, RMSE = 0.171, and r = 0.776** on chronologically held-out test data.
  - Developed an **AI-Physics hybrid framework** by integrating **physics-derived prior features** into temporal ML models, combining process knowledge with data-driven learning for improved f_w estimation.
- **HAQAGen: NIR-to-RGB Spectral Image Colorization** (BS Thesis) *Guide: Dr. Samiran Das, (Jan – May 2025)*
  - Designed **HAQAGen**, a NIR-to-RGB colorization framework that extends a **Mamba**-based (ColorMamba) backbone with a **dual-branch RGB–HSV generator** and **SPADE**-conditioned decoding for improved local chromatic consistency.
  - Formulated a composite objective combining a **differentiable histogram (CDF) loss**, **VGG-19 perceptual loss**, and **frozen-autoencoder texture supervision** to enforce global chromatic consistency and preserve fine-grained textures.
  - Engineered a **resolution-invariant sliding-window inference** pipeline with Hanning blending for seam-free native resolution processing, achieving **24.96 dB PSNR** and up to **34.6% LPIPS** / **23.3% AE** reduction across 4 benchmarks.

Publications

- **Histogram Assisted Quality Aware Generative Model for Resolution Invariant NIR Image Colorization**, Accepted at **IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2026** (Main Author)

Projects

- **Gradewise AI: Intelligent Automated Grading System** (Self Project) *(Apr 2026)*
  - Built an evidence-grounded grading pipeline that decomposes questions into atomic rubrics and combines **LLM, DeBERTa-v3, and Sentence-MPNet** for consensus-based character-level span extraction.
  - Engineered a **confidence-gated scoring heuristic** using span Jaccard overlap, score variance, and lexical grounding; achieved **0.8620 QWK** with automated **Human-in-the-Loop (HITL)** routing for uncertain submissions.
- **TripMate-AI: Agentic Travel Planner** (Self Project) *(Mar 2026)*
  - Built a **LangGraph multi-agent travel planner** with Supervisor, Guardrails, and specialized agents for transit, hotels, weather, and budget-aware itinerary generation with dynamic destination discovery and recommendation.
  - Implemented **stateful MCP tool orchestration** with parallel agent execution, structured outputs, **multi-provider LLM fallbacks**, deterministic preference-based ranking, and **Human-in-the-Loop** iterative plan refinement.
- **India Heatwave Forecasting** (Course Project: KSS605) *(Feb – Apr 2026)*
  - Developed a **spatiotemporal ConvLSTM** for 5-day heatwave forecasting across India using **12 meteorological and radiative channels** on a 33×31 grid, with **leakage-safe P95 labeling** and heatwave-weighted MSE.
  - Evaluated forecasting and heatwave detection using **RMSE, MAE, R^2, CSI, POD, FAR** and lead-time degradation; achieved **1.20°C RMSE**, **0.954 R^2**, and **0.652 POD**, with occlusion-based XAI for model interpretation.
- **Cyclone Track Prediction** (Course Project: EE708) *(Sept – Nov 2025)*
  - Developed a **multi-horizon cyclone trajectory forecasting system** using **149 physics-informed spatio-temporal features** and coordinate-displacement targets to predict cyclone positions at **6–72h** lead times.
  - Engineered **kinematic, intensity, rotational, lagged, and multi-scale temporal features**, including Coriolis dynamics, angular momentum, velocity trends, pressure gradients, and distance-to-land for trajectory modeling.
  - Benchmarked **7 ML models and ensemble learning** under a strict 1980–2019 train / 2020–2025 test split; achieved **11.09 km RMSE at 6h** and **60.87 km MAE at 24h**, reducing RMSE by up to **40.9%** over Ridge.

Technical Skills & Relevant Coursework

- **Programming Languages:** Python, C, SQL
- **Machine Learning & DL:** PyTorch, TensorFlow, Scikit-Learn, CNN, RNN, LSTM, Transformers
- **Generative AI & Agents:** LangChain, LangGraph, AI Agents, RAG, LLM Fine-Tuning, MCP, QLoRA, Vector DBs
- **Data Science/ Libraries:** NumPy, Pandas, Matplotlib, Seaborn, GeoPandas, Feature Engineering, Time-Series Modeling
- **Tools & Deployment:** Git/GitHub, Linux, Docker, FastAPI, Streamlit, Hugging Face, LaTeX
- **Mathematics:** Linear Algebra, Multivariable Calculus, Probability & Statistics
- **Relevant Coursework:** ML, Deep Learning, Computer Vision, AI, Data Structures and Algorithms, DBMS